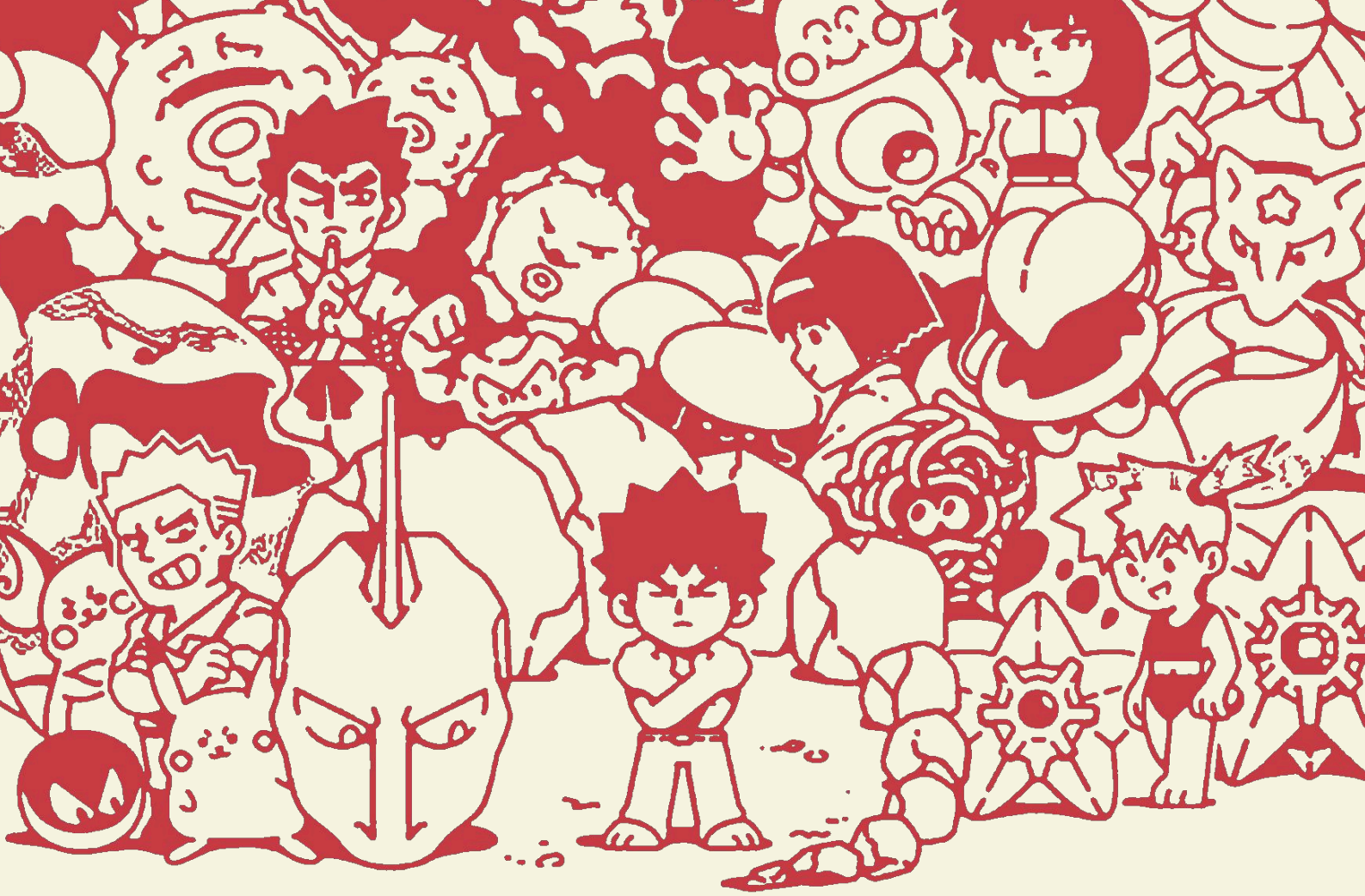
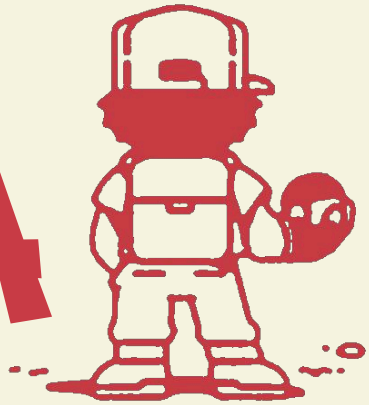




GOTTA CODE EM ALL

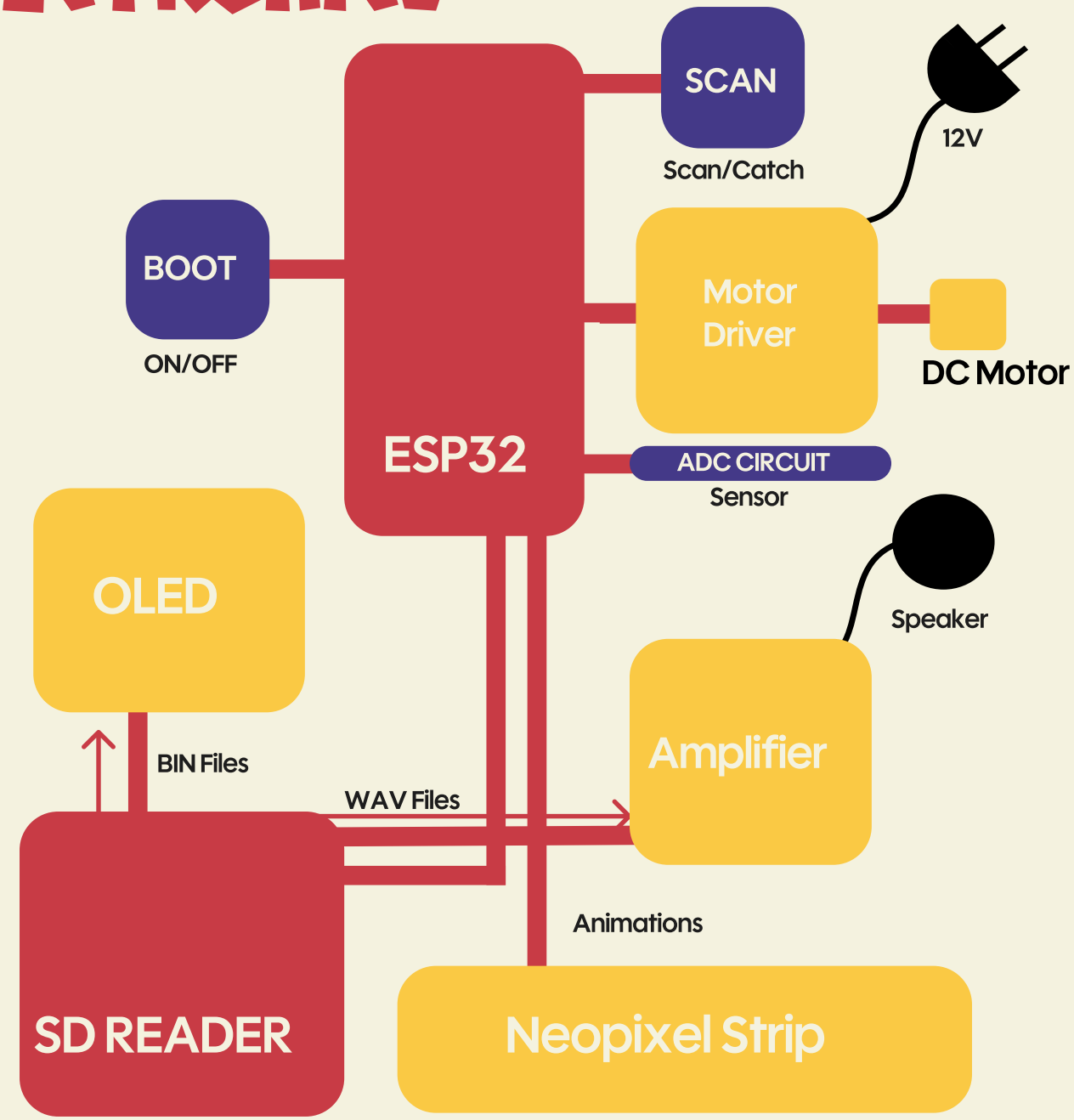
ODT CAPSTONE PROJECT

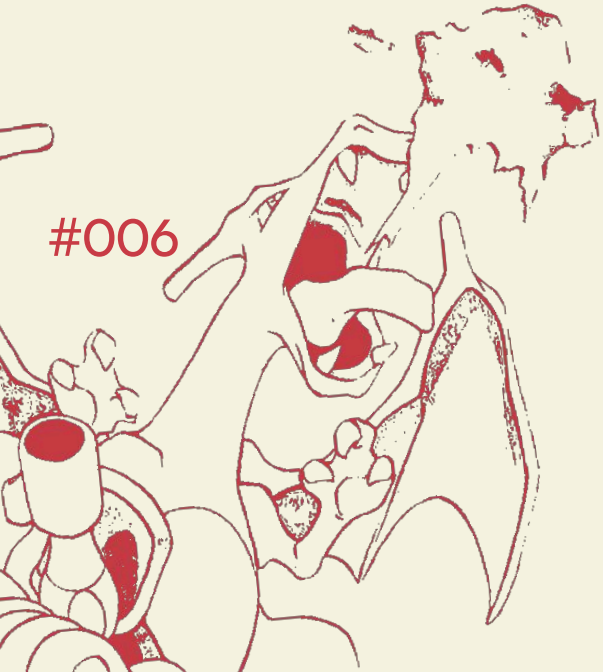
Reflections

Sachit Patnaik
Tanay Dhavate

BITS Design
2025-29

ELECTRONICS



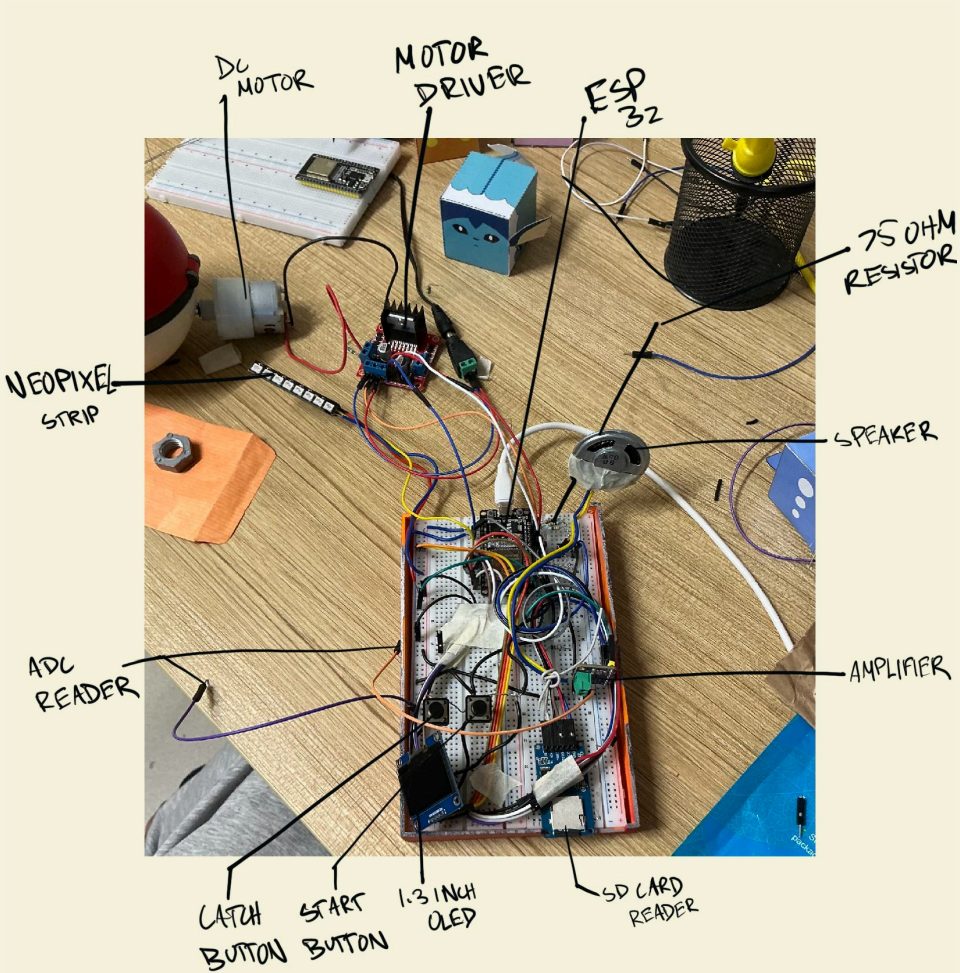


The block diagram shows an ESP32-based Pokédex where the ESP32 controls all system functions, handling inputs from the BOOT and SCAN buttons and identifying Pokémon through an analog sensor. It outputs visuals to an OLED display and Neopixel LEDs, plays audio through an I2S amplifier and speaker, and drives a motor for haptic feedback during captures. Data for animations and sounds is stored on a microSD card via SPI, and the system is powered through a regulated supply, combining multiple modules into a single interactive device.

COMPONENT TABLE

	No.	Purpose	Justification
ESP 32 Microcontroller	1	Main microcontroller that runs the entire system (code execution, I/O control)	Handles I2C (OLED), SPI (SD card), I2S (audio), ADC (sensor), and GPIO simultaneously. Powerful enough for animations + audio playback + logic. Built-in WiFi/Bluetooth (not used here, but future-expandable)
Micro SD Card	1	Stores audio files and animation frames as .bin and .wav files	Esp internal flash memory would be too small to hold all the files needed. We also wanted one single place to store all the functions to be easily called upon later
1.3in OLED Display (SH1106)	1	Displays Pokemon animation sequence and stats. Also shows text instructions	Provided key visual data about pokemon. Screen is also I2C, only 2 pins needed to connect. Monochrome screen provided a good contrast and retro aesthetic
Neopixel LED Strip(8 LEDs)	1	Provide a visual que for key events through the process(scanning, catching, caught/not caught, Type)	We needed a way to show bootup, catch and type animations without using the screen, chose a strip over a ring because of its smaller profile and it fit with our aesthetic
I2S Audio Amplifier	1	Convert electric signals into readable sound waves for the speaker	Needed to regulate sound playback and allow the speaker to work because the ESP outputs audio digitally, not analog
Speaker	1	Play a short description of each Pokemon after each scan and sound effects for the pokeball	Wanted to go for a analog, retro aesthetic which was achieved by the speakers sound quality

	No.	Purpose	Justification
Motor Driver Module	1	Controls the direction and power supplied to the DC motor	ESP32 cannot drive motors directly due to current limitations, allows forward and reverse motion for shake effects . Protects the microcontroller from damage
DC Motor	1	Create the effect of a pokeball catch animation by shaking it side to side	Creates haptic feedback to the user and helps complete the entire system flow. Used a DC motor over a servo due to technical setbacks while 3D printing the Pokeball model
Push Buttons	2	BOOT and SCAN buttons to activate whole system and trigger scan and catch mechanisms	Simple inputs that can be read easily by the ESP32. Clear use case for users, can also have dual functions with long press
Micro SD Card Reader	1	Interfaces the microSD card with the ESP32 to enable reading of stored data	Uses SPI protocol for reliable and relatively fast communication. Allows the ESP32 to access external storage beyond its internal memory, required for streaming animation frames and audio files in real time



LABELED CIRCUIT